Type MS · Explain how ADH changes urine volume and concentration

Answer 1 of 9 · Question 6 part c · 5 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

Number				
6	(a)	L = Bowman's capsule; M = glomerulus; N = collecting duct; 0 = loop of Henlé ;		4
	(b) (i)	• ultrafiltration; • high blood pressure in B/glomerulus; • forces fluid/water into L/Bowman's capsule/out;		3
	(ii)	• blood cells/named; • proteins;		2
	(iii)	line labelled G pointing to proximal convoluted tubule;		1
	(c)	any five from • more sweat; • hypothalamus detects decreased water potential of blood; • ADH secreted; • increases permeability of collecting duct; • causes more reabsorption of water; • volume of filtrate reduced; • concentration of filtrate/urine increased;		5

Total 15 marks

Type MS · Explain how ADH changes urine volume and concentration

Answer 1 of 9 · Question 6 part c · 5 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how ADH changes urine volume and concentration

Answer 2 of 9 · Question 7 part a.i · 1 mark

2023 JAN · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

Number					
7	(a)	(i)	Hypothalamus/osmoregulatory centre;		1
		(ii)	<table><tr><th></th><th colspan="3">Structure</th></tr><tr><th>Function</th><th>glomerulus</th><th>proximal tubule</th><th>renal vein</th></tr><tr><td>regulation of water level</td><td></td><td></td><td></td></tr><tr><td>ultrafiltration</td><td>✓</td><td></td><td></td></tr><tr><td>reabsorption of glucose</td><td></td><td>✓</td><td></td></tr><tr><td>removal of blood from kidney</td><td></td><td></td><td>✓</td></tr><tr><td>supply of blood to kidney</td><td></td><td></td><td></td></tr><tr><td>removal of urine from collecting duct</td><td></td><td></td><td></td></tr></table>	Reject more than one tick in any column	3
	(b)	(i)	• breakdown of proteins/amino acids/deamination/removal of NH_2/amine group; • in the liver;		2
		(ii)	Five from: • second sample has less water/more urea/other components; • urine is more concentrated/less urine produced; • person is less hydrated/less water/lower water potential in the blood; • ADH has been produced; • increased the permeability of the collecting duct; • more water has been reabsorbed (by the blood);		5

Total 11 marks

Type MS · Explain how ADH changes urine volume and concentration

Answer 2 of 9 · Question 7 part a.i · 1 mark

2023 JAN · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how ADH changes urine volume and concentration

Answer 3 of 9 · Question 6 part c · 2 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

6	(a)	(i)	A; (brain)		1
		(ii)	B; (endocrine)		1
	(b)	(i)	• 75 (from weight of man) (1) • 320 × 2 (extra from drinking beer) (1) • 715cm^3 (1)		3
		(ii)	• 715 - 75 (1) • (640 ÷ 75) × 100 (1) • = 853% (1)		3
	(c)		any two from • increases permeability of collecting duct (1) • more water reabsorbed (1) • into blood (1)		2
	(d)		any two from • red blood cells loses water (1) • become distorted (1) • less oxygen transported (1) • volume of plasma decreases (1) • blood becomes thicker/more difficult to circulate(1) • increase in blood pressure (1)	Ignore less water in blood Allow water potential decreases	2
	(e)		any three from • waste products continue to be produced (1) • toxic (1) • need to be removed in urine (1) • less reabsorption of water (1) • less ADH produced (1)	Allow named waste product(s)	3

Total 15 marks

Type MS · Explain how ADH changes urine volume and concentration

Answer 3 of 9 · Question 6 part c · 2 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks
		Allow reverse for whole	

Type MS · Explain how ADH changes urine volume and concentration

Answer 4 of 9 · Question 5 part c · 4 marks

2022 JUN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

5 (a)	• collect urine throughout the cold day; • measure the volume produced; • repeat for a hot day; • compare values;		4
(b)	any three from • vasodilation/arteriole widens; • increased flow of blood through skin; • heat lost through the skin's (surface); • by radiation;	allow artery	3
(c)	• (more) ADH released; • from pituitary gland; • makes kidney tubules/collecting ducts (more) permeable; • (more) water reabsorbed into the body/blood/from filtrate/urine;		4
			Total 11

Type MS · Explain how ADH changes urine volume and concentration

Answer 4 of 9 · Question 5 part c · 4 marks

2022 JUN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how ADH changes urine volume and concentration

Answer 5 of 9 · Question 7 part b.ii · 5 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

Number					
7	(a)	(i)	Hypothalamus/osmoregulatory centre;		1
		(ii)	<table><tr><th></th><th colspan="3">Structure</th></tr><tr><th>Function</th><th>glomerulus</th><th>proximal tubule</th><th>renal vein</th></tr><tr><td>regulation of water level</td><td></td><td></td><td></td></tr><tr><td>ultrafiltration</td><td>✓</td><td></td><td></td></tr><tr><td>reabsorption of glucose</td><td></td><td>✓</td><td></td></tr><tr><td>removal of blood from kidney</td><td></td><td></td><td>✓</td></tr><tr><td>supply of blood to kidney</td><td></td><td></td><td></td></tr><tr><td>removal of urine from collecting duct</td><td></td><td></td><td></td></tr></table>	Reject more than one tick in any column	3
	(b)	(i)	• breakdown of proteins/amino acids/deamination/removal of NH_2/amine group; • in the liver;		2
		(ii)	Five from: • second sample has less water/more urea/other components; • urine is more concentrated/less urine produced; • person is less hydrated/less water/lower water potential in the blood; • ADH has been produced; • increased the permeability of the collecting duct; • more water has been reabsorbed (by the blood);		5

Total 11 marks

Type MS · Explain how ADH changes urine volume and concentration

Answer 5 of 9 · Question 7 part b.ii · 5 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how ADH changes urine volume and concentration

Answer 6 of 9 · Question 9 part whole · 5 marks

2024 JUN · Paper 01 (regional R) · find part whole in this mark scheme extract

9	Any 5 from • (Change in the water) potential/concentration/water level of blood detected by osmoreceptors/hypothalamus; • (when water potential/eq of blood) is low the pituitary gland releases/produces more ADH; • ADH travels through the blood/plasma; • ADH makes the kidney tubules/collecting duct more permeable; • More water reabsorbed/less water absorbed into the blood; • Production of small volume/concentrated urine;	Accept converse for mps 2,4,5 and 6	(5)

Total for Question 9 = 5 marks

Type MS · Explain how ADH changes urine volume and concentration

Answer 7 of 9 · Question 6 part c · 5 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

6	(a)	(i)	any two from: • alcohol absorbed/passes into blood; • rapidly/immediately/straight away/steep increase; • by stomach;	ignore intestines	2
		(ii)	• (alcohol carried to) liver; • broken down/detoxified; • level (in blood) falls;		3
	(b)	(i)	• any time after (including) 2.4 hours/2 hours 24 mins;	allow a time of between 1-20 minutes	1
		(ii)	• alcohol is a depressant; • slows reaction/increases reaction time; • drivers can't respond as quickly/less alert; • leads to accidents;		4
	(c)		• collecting duct less permeable (in absence of ADH); • less water reabsorbed; • into blood; • volume of urine (therefore) increased/more urine produced; • and is more dilute/less concentrated;	accept ref fewer aquaporins/water channels	5
					15 marks

Type MS · Explain how ADH changes urine volume and concentration

Answer 7 of 9 · Question 6 part c · 5 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how ADH changes urine volume and concentration

Answer 8 of 9 · Question 1 part b.iii · 3 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part b.iii in this mark scheme extract

Number					
1	(a)	(i)	• first convoluted tubule; • loop of Henlé; • second convoluted tubule;	diagram only, no labels required	3
		(ii)	44mm; 44 ÷ 650; = 0.068mm;	ecf from mp1 max 2 marks Allow 0.0676	3
		(iii)	• protein; • red/white blood cells/platelets;		2
		(iv)	glucose;		1
	(b)	(i)	Any 2 from: • same person; • same activity levels; • same volume of water; • same food eaten;		2
		(ii)	710 ÷ 4; = 178 cm^3 per hour;	Allow 1 mark if division by 4 is shown Allow 177.5	2
		(iii)	• less urine production/lower urine loss on day two; • more ADH secreted on second day; • more water reabsorbed; • more water lost by sweating;	Allow reverse argument for all marking points	3

Total 16 marks

Type MS · Explain how ADH changes urine volume and concentration

Answer 8 of 9 · Question 1 part b.iii · 3 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part b.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain how ADH changes urine volume and concentration

Answer 9 of 9 · Question 6 part e · 3 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part e in this mark scheme extract

6	(a)	(i)	A; (brain)		1
		(ii)	B; (endocrine)		1
	(b)	(i)	• 75 (from weight of man) (1) • 320 × 2 (extra from drinking beer) (1) • 715cm^3 (1)		3
		(ii)	• 715 - 75 (1) • (640 ÷ 75) × 100 (1) • = 853% (1)		3
	(c)		any two from • increases permeability of collecting duct (1) • more water reabsorbed (1) • into blood (1)		2
	(d)		any two from • red blood cells loses water (1) • become distorted (1) • less oxygen transported (1) • volume of plasma decreases (1) • blood becomes thicker/more difficult to circulate(1) • increase in blood pressure (1)	Ignore less water in blood Allow water potential decreases	2
	(e)		any three from • waste products continue to be produced (1) • toxic (1) • need to be removed in urine (1) • less reabsorption of water (1) • less ADH produced (1)	Allow named waste product(s)	3

Total 15 marks

Type MS · Explain how ADH changes urine volume and concentration

Answer 9 of 9 · Question 6 part e · 3 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part e in this mark scheme extract

Question number	Answer	Notes	Marks
		Allow reverse for whole	